

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Sol :The correct OSI Model sequence is:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Missing Layers:

- 1. Session Layer**
- 2. Data Link Layer**

Explanation:

1. Session Layer

- Establishes, manages, and terminates communication sessions between devices.
- Keeps data exchange organized and synchronized.
- Helps recover communication if the connection is interrupted.

2. Data Link Layer

- Ensures error-free data transfer between two directly connected devices.
- Detects and corrects transmission errors.
- Controls data flow and uses MAC addresses for device-to-device communication.

Why They Are Essential:

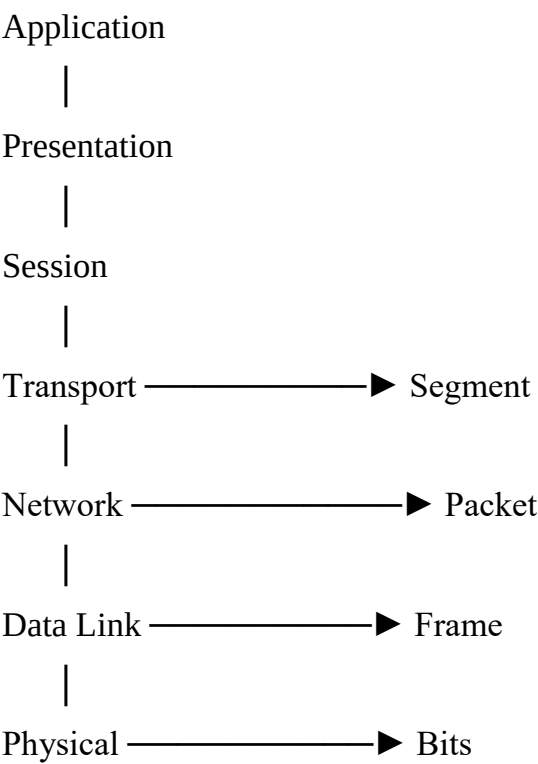
- The Session Layer maintains a stable communication session, ensuring continuous and organized data exchange.

- The Data Link Layer provides reliable transmission by detecting errors and delivering data correctly over the physical network.

Final **Answer:**
Application → Presentation → Session → Transport → Network → Data Link → Physical

3. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Sol:



Analysis of Data Unit Transformation

OSI Layer	Data Unit	Function
Transport	Segment	Divides data into smaller segments and provides reliable delivery using error and flow control.
Network	Packet	Adds IP addresses and selects the best path for data transmission.
Data Link	Frame	Adds MAC addresses and error detection to ensure reliable communication between devices.
Physical	Bits	Converts frames into binary bits (0s and 1s) and transmits them through the communication medium.

How Transformation Supports Transmission

- **Data → Segment:** The Transport layer breaks large data into manageable segments.
- **Segment → Packet:** The Network layer adds logical (IP) addressing for routing.

- **Packet → Frame:** The Data Link layer adds physical (MAC) addresses and error-checking information.
- **Frame → Bits:** The Physical layer converts frames into binary bits for transmission over cables, fiber, or wireless signals.

Conclusion:

The transformation of **Segment → Packet → Frame → Bits** ensures that data is correctly addressed, routed, error-checked, and transmitted reliably from the sender to the receiver.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

Sol: **Concept Map**

Data Link Layer → MAC Address → Local Delivery

Network Layer → IP Address → Global Delivery

Explanation

- **Data Link Layer → MAC Address → Local Delivery**
 - Uses **MAC addresses** to identify devices on the same local network (LAN).
 - Delivers data correctly between directly connected devices.
- **Network Layer → IP Address → Global Delivery**
 - Uses **IP addresses** to identify devices across different networks.
 - Routes packets through routers to reach the destination anywhere on the Internet.

How These Relationships Ensure Complete Data Delivery

- The **IP address** identifies the destination across multiple networks and finds the best route.
- The **MAC address** ensures the data is delivered to the correct device on each local network.
- Routers use **IP addresses** for routing, while switches use **MAC addresses** for local forwarding.
- Together, **IP provides global delivery** and **MAC provides local delivery**, ensuring data reaches the correct destination accurately and efficiently.

Conclusion:

The **Network Layer (IP Address)** handles **global routing**, while the **Data Link Layer**

(MAC Address) handles **local device delivery**. Their combined operation ensures complete and reliable data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Sol:

Error Control

|



Data Link Layer —→▶ Error Detection

|



Transport Layer —→▶ Error Correction

Explanation

- **Data Link Layer → Error Detection**
 - Detects errors in frames using methods like **CRC (Cyclic Redundancy Check)** or checksums.
 - Discards corrupted frames or requests retransmission over the local link.
- **Transport Layer → Error Correction**
 - Ensures end-to-end reliable communication.
 - Uses **acknowledgments (ACKs), sequence numbers, and retransmissions** to recover lost or damaged data.

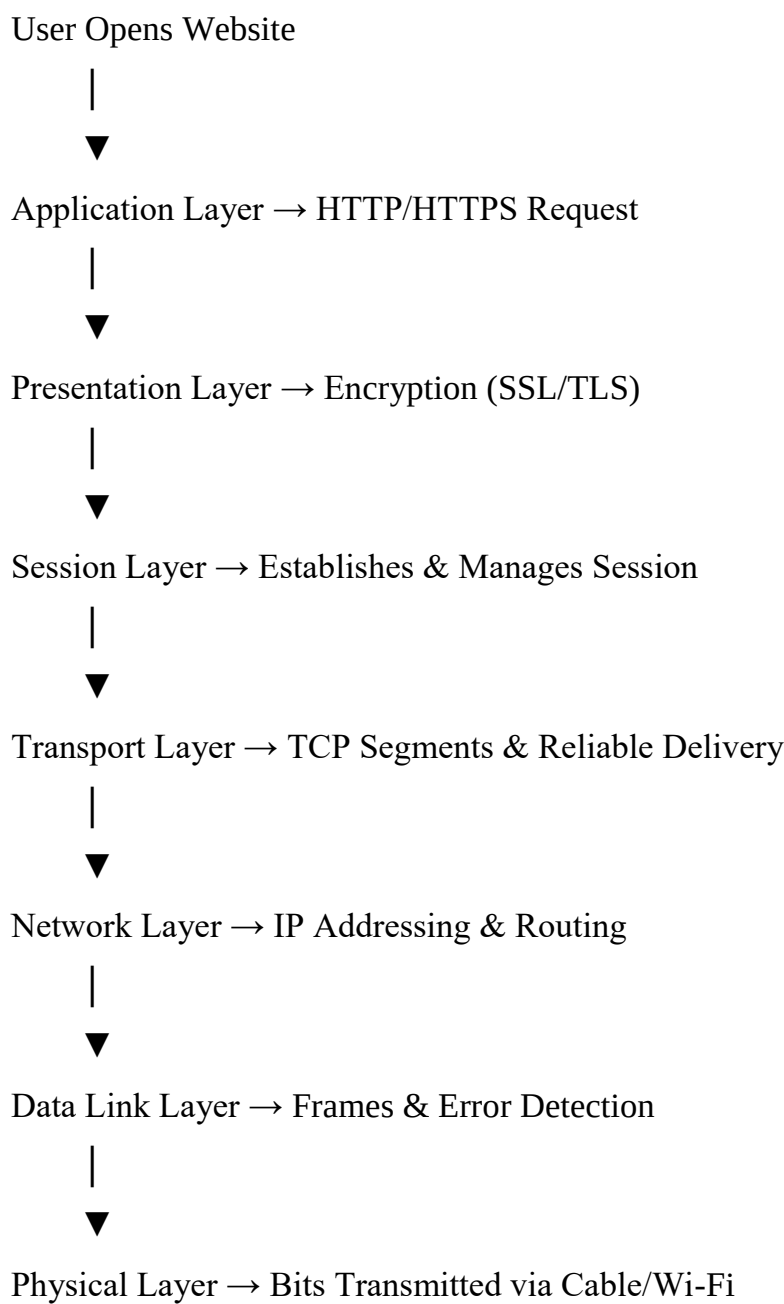
Analysis

- The **Data Link Layer** detects transmission errors between directly connected devices.
- The **Transport Layer** corrects errors by retransmitting lost or corrupted segments and delivering data in the correct order.
- Together, these layers ensure **accurate, reliable, and complete data transmission** from sender to receiver.

Conclusion:

The interaction of the **Data Link Layer (error detection)** and the **Transport Layer (error correction)** improves communication reliability by detecting errors early and ensuring that the receiver gets complete and error-free data.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.



6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

Concept Map

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analysis

- **Physical Layer ✓:** The physical connection (cables, Wi-Fi, signals) is working correctly.
- **Data Link Layer ✓:** Data frames are being transmitted successfully on the local network.
- **Network Layer ✗:** The problem starts here. Possible causes include incorrect IP address, routing failure, or router malfunction.

- **Transport Layer ✕:** Since the Network layer fails, TCP/UDP cannot establish communication.
- **Application Layer ✕:** Applications (such as a web browser or email) cannot access network services because lower layers have failed.

How the Concept Map Helps in Troubleshooting

- It shows that the **Physical** and **Data Link** layers are functioning correctly.
- It identifies the **Network layer** as the first point of failure.
- Troubleshooting should begin at the **Network layer** by checking:
 - IP address configuration
 - Default gateway
 - Router status
 - Routing tables
- Once the Network layer issue is fixed, the **Transport** and **Application** layers should start working automatically.

Conclusion

The concept map helps locate the exact layer where communication fails. Since the first failure occurs at the **Network layer**, technicians can focus on IP addressing and routing instead of checking the physical connection, making troubleshooting faster and more efficient.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation